Lab 5 practice: variation of parameters (Zill §3.5)

Prep for the Wed Oct 14 lab quiz. Closed book, about 10 minutes per problem. Solve each equation, using variation of parameters unless the problem says otherwise. Problems 1 to 5 have forcing that undetermined coefficients cannot handle, problem 6 is variable-coefficient, problem 7 is an initial value problem, and problem 8 can be done both ways. Every solution was checked by substituting back (`lab5_check.py`).

Standard form first. Divide through so the equation reads $y'' + P(x)y' + Q(x)y = f(x)$. The f in every formula below is the right side of *this* form.

Step 1. Solve the homogeneous equation: $y_c = c_1 y_1 + c_2 y_2$. For constant coefficients, $am^2 + bm + c = 0$ gives:

Roots	$y_1,\ y_2$
distinct real m_1, m_2	$e^{m_1 x},\ e^{m_2 x}$
repeated real m	$e^{mx},\ xe^{mx}$
complex $\alpha \pm \beta i$	$e^{\alpha x}\cos\beta x,\ e^{\alpha x}\sin\beta x$

Step 2. Wronskian and its two companions:

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1', \qquad W_1 = \begin{vmatrix} 0 & y_2 \\ f & y_2' \end{vmatrix} = -y_2 f, \qquad W_2 = \begin{vmatrix} y_1 & 0 \\ y_1' & f \end{vmatrix} = y_1 f$$

Step 3. $u_1' = \dfrac{W_1}{W}$ and $u_2' = \dfrac{W_2}{W}$. Integrate for u_1 and u_2 (the constants of integration can be dropped).

Step 4. $y_p = u_1 y_1 + u_2 y_2$ and $y = y_c + y_p$. Apply initial conditions last, to the whole y.

Problem 1

$y'' + y = \sec x$

Solution: constant coefficients, forcing outside the table

Second order, linear, constant coefficients. The forcing $\sec x$ is not a polynomial, exponential, sine or cosine, and its derivatives never close into a finite family, so undetermined coefficients fails. Use variation of parameters.

Already in standard form, with $f = \sec x$. $m^2 + 1 = 0$ gives $m = \pm i$, so $y_1 = \cos x$, $y_2 = \sin x$, and

$$y_c = c_1 \cos x + c_2 \sin x$$

$$W = \cos x \cdot \cos x - (-\sin x) \sin x = 1$$

$$W_1 = -\sin x \sec x = -\tan x, \qquad W_2 = \cos x \sec x = 1$$

$$u_1' = -\tan x \implies u_1 = \ln|\cos x|, \qquad u_2' = 1 \implies u_2 = x$$

On the interval below $\cos x > 0$, so $u_1 = \ln(\cos x)$. Then

$$y_p = \cos x \, \ln(\cos x) + x \sin x$$

$$\boxed{y = c_1 \cos x + c_2 \sin x + \cos x \, \ln(\cos x) + x \sin x, \qquad -\tfrac{\pi}{2} < x < \tfrac{\pi}{2}}$$

The interval comes from $f = \sec x$, which blows up at $x = \pm\pi/2$. The same formula works on any other interval between consecutive zeros of $\cos x$, with $\ln|\cos x|$ in place of $\ln(\cos x)$.

Problem 2

$y'' + y = \tan x$

Solution: same homogeneous part, trickier integral

Same left side as problem 1, so $y_c = c_1 \cos x + c_2 \sin x$ and $W = 1$. Here $f = \tan x$.

$$W_1 = -\sin x \tan x = -\frac{\sin^2 x}{\cos x} = \cos x - \sec x, \qquad W_2 = \cos x \tan x = \sin x$$

$$u_1' = \cos x - \sec x \implies u_1 = \sin x - \ln|\sec x + \tan x|$$

$$u_2' = \sin x \implies u_2 = -\cos x$$

The step $\sin^2 x / \cos x = \sec x - \cos x$ comes from writing $\sin^2 x = 1 - \cos^2 x$. Now assemble:

$$y_p = \cos x(\sin x - \ln|\sec x + \tan x|) - \cos x \sin x = -\cos x \ln|\sec x + \tan x|$$

The $\sin x \cos x$ terms cancel.

$$\boxed{y = c_1 \cos x + c_2 \sin x - \cos x \ln|\sec x + \tan x|, \qquad -\tfrac{\pi}{2} < x < \tfrac{\pi}{2}}$$

On this interval $\sec x + \tan x > 0$, so the absolute value can be dropped. The interval is the same as in problem 1, because $\tan x$ also fails at $\pm\pi/2$.

Problem 3

$y'' + 9y = \csc 3x$

Solution: frequency 3, so the Wronskian is not 1

$m^2 + 9 = 0$ gives $m = \pm 3i$, so $y_1 = \cos 3x$, $y_2 = \sin 3x$, $f = \csc 3x$.

$$W = \cos 3x \cdot 3\cos 3x - (-3\sin 3x)\sin 3x = 3$$

$$W_1 = -\sin 3x \csc 3x = -1, \qquad W_2 = \cos 3x \csc 3x = \cot 3x$$

$$u_1' = -\frac{1}{3} \implies u_1 = -\frac{x}{3}, \qquad u_2' = \frac{\cot 3x}{3} \implies u_2 = \frac{1}{9}\ln|\sin 3x|$$

The factor $\frac{1}{3}$ in u_2 comes from $\int \cot 3x\, dx = \frac{1}{3}\ln|\sin 3x|$, on top of the $\frac{1}{3}$ from dividing by W.

$$y_p = -\frac{x}{3}\cos 3x + \frac{1}{9}\sin 3x\, \ln|\sin 3x|$$

$$\boxed{y = c_1 \cos 3x + c_2 \sin 3x - \frac{x}{3}\cos 3x + \frac{1}{9}\sin 3x\, \ln(\sin 3x), \qquad 0 < x < \tfrac{\pi}{3}}$$

$\csc 3x$ is undefined where $\sin 3x = 0$, that is at $x = 0, \pm\pi/3, \ldots$. The interval $0 < x < \pi/3$ has $\sin 3x > 0$, so the absolute value is gone.

Problem 4

$y'' - 2y' + y = \dfrac{e^x}{x}$

Solution: repeated root, forcing e^x/x

$m^2 - 2m + 1 = (m-1)^2 = 0$, a repeated root. So $y_1 = e^x$, $y_2 = xe^x$, and $f = e^x/x$ (the leading coefficient is already 1).

$$W = e^x(e^x + xe^x) - e^x \cdot xe^x = e^{2x}$$

$$W_1 = -xe^x \cdot \frac{e^x}{x} = -e^{2x}, \qquad W_2 = e^x \cdot \frac{e^x}{x} = \frac{e^{2x}}{x}$$

$$u_1' = -1 \implies u_1 = -x, \qquad u_2' = \frac{1}{x} \implies u_2 = \ln x$$

$$y_p = -xe^x + xe^x \ln x$$

The term $-xe^x$ is a multiple of y_2, so it can be absorbed into c_2. Keeping it is also correct.

$$\boxed{y = c_1 e^x + c_2 xe^x + xe^x \ln x, \qquad x > 0}$$

$f = e^x/x$ is undefined at $x = 0$, and $\ln x$ needs $x > 0$. The $\ln x$ in the answer is also why undetermined coefficients could never guess it: no entry in the table produces a logarithm.

Problem 5

$3y'' - 6y' + 6y = e^x \sec x$

Solution: leading coefficient is not 1

The leading coefficient is 3, so **divide first**. The standard form is

$$y'' - 2y' + 2y = \tfrac{1}{3}e^x \sec x, \qquad f = \tfrac{1}{3}e^x \sec x$$

$m^2 - 2m + 2 = 0$ gives $m = 1 \pm i$, so $y_1 = e^x \cos x$, $y_2 = e^x \sin x$.

$$W = e^x \cos x\,(e^x \sin x + e^x \cos x) - (e^x \cos x - e^x \sin x)\,e^x \sin x = e^{2x}$$

$$W_1 = -e^x \sin x \cdot \tfrac{1}{3}e^x \sec x = -\tfrac{1}{3}e^{2x} \tan x, \qquad W_2 = e^x \cos x \cdot \tfrac{1}{3}e^x \sec x = \tfrac{1}{3}e^{2x}$$

$$u_1' = -\tfrac{1}{3}\tan x \implies u_1 = \tfrac{1}{3}\ln(\cos x), \qquad u_2' = \tfrac{1}{3} \implies u_2 = \tfrac{x}{3}$$

$$y_p = \tfrac{1}{3}e^x \cos x\, \ln(\cos x) + \tfrac{1}{3}xe^x \sin x$$

$$\boxed{y = e^x(c_1 \cos x + c_2 \sin x) + \tfrac{1}{3}e^x\big[\cos x\, \ln(\cos x) + x \sin x\big], \qquad -\tfrac{\pi}{2} < x < \tfrac{\pi}{2}}$$

If you skip the division and use $f = e^x \sec x$, every u comes out three times too large. That is the standard mistake for this type of problem.

Problem 6

$x^2y'' + xy' - y = \ln x$, given that $y_1 = x$ and $y_2 = x^{-1}$ solve the homogeneous equation.

Solution: variable coefficients, standard form is the whole trick

Second order, linear, variable coefficients (Cauchy–Euler). Undetermined coefficients does not apply. The two homogeneous solutions are supplied. Quick check: for $y = x$, $0 + x - x = 0$; for $y = 1/x$, $x^2 \cdot 2/x^3 - x/x^2 - 1/x = 0$. Both hold, so

$$y_c = c_1 x + c_2 x^{-1}$$

Divide by x^2. The standard form is $y'' + \frac{1}{x}y' - \frac{1}{x^2}y = \dfrac{\ln x}{x^2}$, so $f = \dfrac{\ln x}{x^2}$, not $\ln x$.

$$W = x\Big(-\frac{1}{x^2}\Big) - \frac{1}{x}(1) = -\frac{2}{x}$$

$$W_1 = -\frac{1}{x}\cdot\frac{\ln x}{x^2} = -\frac{\ln x}{x^3}, \qquad W_2 = x\cdot\frac{\ln x}{x^2} = \frac{\ln x}{x}$$

$$u_1' = \frac{-\ln x/x^3}{-2/x} = \frac{\ln x}{2x^2}, \qquad u_2' = \frac{\ln x/x}{-2/x} = -\frac{\ln x}{2}$$

Integrate by parts, using $\int x^{-2}\ln x\,dx = -\dfrac{\ln x + 1}{x}$ and $\int \ln x\,dx = x\ln x - x$:

$$u_1 = -\frac{\ln x + 1}{2x}, \qquad u_2 = -\frac{x\ln x - x}{2}$$

$$y_p = x\Big(-\frac{\ln x + 1}{2x}\Big) + \frac{1}{x}\Big(-\frac{x\ln x - x}{2}\Big) = -\frac{\ln x + 1}{2} - \frac{\ln x - 1}{2} = -\ln x$$

$$\boxed{y = c_1 x + c_2 x^{-1} - \ln x, \qquad x > 0}$$

Check by hand: $y_p = -\ln x$ has $y_p' = -1/x$ and $y_p'' = 1/x^2$, so $x^2 y_p'' + x y_p' - y_p = 1 - 1 + \ln x = \ln x$.

Problem 7

$y'' + 2y' + y = e^{-x} \ln x, \qquad y(1) = 0, \quad y'(1) = e^{-1}$

Solution: initial value problem, repeated root

$m^2 + 2m + 1 = (m+1)^2 = 0$, so $y_1 = e^{-x}$, $y_2 = xe^{-x}$, $f = e^{-x}\ln x$. The $\ln x$ factor rules out undetermined coefficients, and the data at $x = 1$ puts the problem on $x > 0$.

$$W = e^{-x}(e^{-x} - xe^{-x}) - (-e^{-x})(xe^{-x}) = e^{-2x}$$

$$W_1 = -xe^{-x}\cdot e^{-x}\ln x = -xe^{-2x}\ln x, \qquad W_2 = e^{-x}\cdot e^{-x}\ln x = e^{-2x}\ln x$$

$$u_1' = -x\ln x, \qquad u_2' = \ln x$$

$$u_1 = -\frac{x^2}{2}\ln x + \frac{x^2}{4}, \qquad u_2 = x\ln x - x$$

$$y_p = e^{-x}u_1 + xe^{-x}u_2 = e^{-x}\Big[-\frac{x^2}{2}\ln x + \frac{x^2}{4} + x^2\ln x - x^2\Big] = e^{-x}\Big[\frac{x^2}{2}\ln x - \frac{3x^2}{4}\Big]$$

General solution first, then the conditions:

$$y = e^{-x}\Big[c_1 + c_2 x + \frac{x^2}{2}\ln x - \frac{3x^2}{4}\Big]$$

$y(1) = e^{-1}[c_1 + c_2 - \frac{3}{4}] = 0$, so $c_1 + c_2 = \frac{3}{4}$. Differentiating,

$$y' = e^{-x}\Big[c_2 + x\ln x + \frac{x}{2} - \frac{3x}{2} - c_1 - c_2 x - \frac{x^2}{2}\ln x + \frac{3x^2}{4}\Big]$$

At $x = 1$ the logarithm vanishes: $y'(1) = e^{-1}[c_2 - 1 - c_1 - c_2 + \frac{3}{4}] = e^{-1}[-c_1 - \frac{1}{4}] = e^{-1}$, so $c_1 = -\frac{5}{4}$ and then $c_2 = 2$.

$$\boxed{y = e^{-x}\Big[-\frac{5}{4} + 2x - \frac{3x^2}{4} + \frac{x^2}{2}\ln x\Big], \qquad x > 0}$$

Problem 8

$y'' - y' - 2y = 6e^{2x}$. Solve it by variation of parameters, then again by undetermined coefficients, and compare.

Solution: the same answer by two routes

Constant coefficients, and the forcing is an exponential, so *either* method works. $m^2 - m - 2 = (m-2)(m+1) = 0$, so $y_1 = e^{2x}$, $y_2 = e^{-x}$ and

$$y_c = c_1 e^{2x} + c_2 e^{-x}$$

Variation of parameters. Here $f = 6e^{2x}$.

$$W = e^{2x}(-e^{-x}) - 2e^{2x}e^{-x} = -3e^{x}$$

$$W_1 = -e^{-x} \cdot 6e^{2x} = -6e^{x}, \qquad W_2 = e^{2x} \cdot 6e^{2x} = 6e^{4x}$$

$$u_1' = \frac{-6e^{x}}{-3e^{x}} = 2 \implies u_1 = 2x, \qquad u_2' = \frac{6e^{4x}}{-3e^{x}} = -2e^{3x} \implies u_2 = -\frac{2}{3}e^{3x}$$

$$y_p = 2xe^{2x} - \frac{2}{3}e^{2x}$$

The second term is a multiple of y_1, so it goes into c_1.

Undetermined coefficients. 2 is a simple root, so the guess Ae^{2x} would vanish on the left. Multiply by x: $y_p = Axe^{2x}$, with $y_p' = A(1+2x)e^{2x}$ and $y_p'' = A(4+4x)e^{2x}$.

$$Ae^{2x}\big[(4+4x) - (1+2x) - 2x\big] = 3Ae^{2x} = 6e^{2x} \implies A = 2$$

Same $y_p = 2xe^{2x}$.

$$\boxed{y = c_1 e^{2x} + c_2 e^{-x} + 2xe^{2x}, \qquad -\infty < x < \infty}$$

Undetermined coefficients took three lines here and variation of parameters took ten, so when f is in the table use that method. Variation of parameters earns its keep when f is not, but it gives the right answer either way, and it is the better check when you are unsure of the overlap rule.